Get Python 3 and Tkinter ready

Use this guide if you want to run the animation on your own Windows, Mac or Linux computer. You do not need to install anything to use Scratch in the browser.

School computer or no install permission? Ask your teacher or IT. Do not bypass device restrictions. Tkinter is not a separate download for a standard Python.org install. **Do not run** `pip install tkinter`.

1 Try the check before you install anything

Open **PowerShell** on Windows or **Terminal** on Mac/Linux. If you see `>>>`, open a fresh terminal. Type only the command after your device name.

```
Windows     python -m tkinter
Mac/Linux   python3 -m tkinter
```

What success looks like: a small Tk demonstration window appears. Close it and go straight to page 2. If Windows already uses the `py` command, you can test with `py -m tkinter` instead.

2 Only if the test did not work: choose your computer

WINDOWS

Go to Python.org Windows downloads. Install the **Python install manager**, then reopen PowerShell. If it has no Python runtime yet, run `py install default`. Run the Tkinter test again. If your older installer offers optional features, keep **tcl/tk and IDLE** selected.

MAC

Open Python.org macOS downloads. Choose a supported stable Python 3 macOS installer. Open the downloaded **.pkg**, follow the installer, reopen Terminal and run the test again.

UBUNTU / DEBIAN

If the system Python 3 is installed but Tkinter is missing, ask an adult or IT to install the python3-tk package through your system package manager. On other Linux distributions, follow your distributor's instructions. Run the test again.

Official help: Tkinter test · Windows install manager · macOS installation

Import Tkinter and run the sprite

Tkinter creates a desktop window. The sprite sheet supplies the pictures. The Python program decides which frame to show and how long to show it.

3 Put both downloads in the same folder

In Main Task 2, download the **sprite sheet PNG** and **animate.py**. Create a folder called **SpriteSheetStudio** in Documents and move both downloads there.

Documents / SpriteSheetStudio

`animate.py`	**1 The program you open and run.**
`sprite_sheet.png`	**2 The image the program loads.**

If the download name ends in **(1)**, check you have the newest image and rename that copy to **sprite_sheet.png** inside your project folder. Keep the file extension.

4 Open the program in IDLE

Open **IDLE**, choose **File > Open** and select **animate.py**. Near the top you will see the import below. It is already in the starter; do not add it again.

```
import tkinter as tk
```

import tkinter loads the window tools. **as tk** gives the module a short name. From the **editor window** (not the Shell), select **Run > Run Module**. The Sprite Sheet Studio window should show your character changing expression.

5 Change one number, then compare

```
DELAY_MS = 400    # 400 milliseconds = 0.4 seconds
```

Start with **400** (set it if yours is different). Close the animation window, change the value to **800**, save and run again. Each picture stays visible twice as long; the order stays the same. Watch two loops. Save the edited **.py** file as evidence.

Stuck? If you see **No module named tkinter**, go back to the test on page 1 and check you used the same Python installation as IDLE. If you see a missing PNG error, check both files are in the same folder and that the image is named exactly **sprite_sheet.png**. If setup is blocked, use Scratch and show your teacher.